

Chapter 1

Exploratory Data Analysis: Summaries and Visual Structure

1.1 Why Summaries and Visual Structure Matter

The previous chapter focused on whether variables are trustworthy enough to analyze. Once we have checked definitions, missingness, units, data types, impossible values, and outliers, we can begin describing what the data appear to say.

Exploratory data analysis, or EDA, is the habit of listening to data before forcing it into a model. This chapter develops two complementary tools:

1. **single-number summary statistics**, which compress one feature of a distribution into one number; and
2. **visual EDA**, which uses plots to reveal distributional shape, relationships, subgroups, and unusual structure.

Good EDA combines both. Numbers make comparisons precise; plots make structure visible. The practical question for this chapter is:

How do we summarize and visualize variables in a way that is accurate, interpretable, and defensible?

1.2 A Numeric Variable as a Vector

Let $x_1, x_2, \dots, x_n$ be the observed values of a numeric variable X for n observations. We can collect these values in a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n. \quad (1.1)$$

This notation is useful because most numerical summaries are functions of the entries of $\mathbf{x}$.

For example, the sample mean can be written as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.2)$$

Using the ones vector $\mathbf{1} \in \mathbb{R}^n$, the same quantity is

$$\bar{x} = \frac{1}{n} \mathbf{1}^T \mathbf{x}. \quad (1.3)$$

This linear algebra view is not required for basic EDA, but it helps us see that summaries are not magic. They are operations applied to the observed data vector.

Math Foundations Connection. A vector is an ordered collection of quantities that pertain to the same object or concept. In EDA, a variable column is a vector of observed values. Dot products, sums, centered vectors, and norms give compact ways to express means, variances, covariances, and correlations. See the Math Foundations textbook, Chapters 2, 3, and 5.

1.3 Single-Number Summary Statistics

A single-number summary statistic characterizes one aspect of a variable's distribution. It does not describe everything about the variable. It answers a specific question.

Summary family	Guiding question	Examples
Center	Where is a typical value?	mean, median, mode
Spread	How variable are the values?	range, IQR, variance, standard deviation
Shape	How is the distribution arranged?	skewness, kurtosis
Extremes	What are the edge cases?	minimum, maximum, outlier flags

Table 1.1: Single-number summaries answer different questions about a variable.

The most common EDA mistake is to compute a statistic without saying what question it answers. The same data set may require several summaries: a center for a typical value, a spread for variability, a shape summary for asymmetry or tails, and an extreme-value check for unusual observations.

1.4 Center and Spread

1.4.1 Measures of Center

Measures of center summarize where a distribution is located. The three most common are the mean, median, and mode.

The **mean** is the balance point:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.4)$$

It uses every value, which makes it natural for many numerical variables, but also sensitive to outliers.

The **median** is the middle value after sorting the data. If

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

denotes the ordered data, then the median is the middle ordered value for odd n and the average of the two middle ordered values for even n . The median is robust to outliers because changing an extreme value usually does not move the middle of the ordered list.

The **mode** is the most common value. It is especially useful for categorical variables. For continuous numerical variables, exact ties may be rare, so analysts often discuss peaks of a histogram or density estimate rather than an exact modal value.

Situation	Useful summary	center	Reason
Symmetric numeric variable	Mean or median		Often similar
Skewed or outlier-prone numerical variable	Median		More robust to tail values
Categorical variable	Mode		Categories do not have arithmetic averages
Ordinal variable	Median or mode		Respects ordering without assuming equal spacing

1.4.2 Measures of Spread

Measures of spread summarize variability. Two variables can have the same center but very different spread.

The **range** is

$$\text{Range}(\mathbf{x}) = \max_i x_i - \min_i x_i. \quad (1.5)$$

It is easy to interpret but depends only on the two most extreme observations.

The **interquartile range** is

$$\text{IQR} = Q_3 - Q_1, \quad (1.6)$$

where Q_1 is the first quartile and Q_3 is the third quartile. Because it measures the spread of the middle half of the data, the IQR is more robust to outliers than the range.

The **sample variance** and **sample standard deviation** are

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad (1.7)$$

$$s_x = \sqrt{s_x^2}. \quad (1.8)$$

Variance is mathematically powerful but is measured in squared units. Standard deviation is often easier to interpret because it is measured in the original units of the variable.

Define the centered data vector

$$\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}. \quad (1.9)$$

Then

$$s_x^2 = \frac{1}{n-1} \tilde{\mathbf{x}}^T \tilde{\mathbf{x}}. \quad (1.10)$$

Thus spread can be interpreted geometrically: large spread means the centered data vector has large squared length.

1.5 Shape and Extremes

Measures of shape describe how a distribution is arranged beyond its center and spread. The two shape ideas emphasized here are **skewness** and **kurtosis**.

For a random variable X with mean μ and standard deviation σ , the population skewness is

$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}. \quad (1.11)$$

Positive skewness indicates a long or heavy right tail; negative skewness indicates a long or heavy left tail. In a right-skewed distribution, the mean is often larger than the median because the right tail pulls the mean upward.

The population kurtosis is

$$\gamma_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \quad (1.12)$$

For EDA, it is safest to interpret kurtosis as tail weight rather than simply as “peakedness.” The normal distribution has kurtosis 3, so excess kurtosis is $\gamma_2 - 3$.

Robust analogs. Mean, variance, skewness, and kurtosis are moment-based summaries; they use arithmetic powers of deviations from the mean. Median, IQR, Bowley skewness, and Moors kurtosis are quantile-based analogs. Quantile-based summaries are often more robust when outliers or heavy tails are present. For this course, the main habit is more important than memorizing every robust formula: when outliers or skewness are visible, compare moment-based summaries with quantile-based summaries.

The simplest measures of extremes are the minimum and maximum. Extremes are not automatically bad: they may be rare but valid observations, data-entry errors, unit mistakes, duplicated records, or evidence of subgroups.

A common outlier flag uses Tukey fences. Values are flagged when they fall outside

$$[Q_1 - 1.5 \text{IQR}, Q_3 + 1.5 \text{IQR}]. \quad (1.13)$$

This rule is tied to the box plot. It is a screening tool, not an automatic deletion rule. A defensible action is: flag the value, inspect the context, and document the decision.

1.6 Visual EDA

Numerical summaries compress. Plots reveal. Visual EDA uses graphics to show the behavior of observations directly. A plot should match the variable types involved.

1.6.1 Univariate Visual EDA

Univariate EDA examines one variable at a time. For categorical or ordinal variables, a **bar plot** displays counts or proportions by category. It helps reveal common levels, rare levels, unexpected labels, and ordering problems.

For numerical variables, a **histogram** groups observations into bins and displays the number or proportion of observations in each bin. A histogram helps inspect symmetry, skewness, modality,

gaps, clusters, outliers, and approximate bell shape. Because histograms depend on bin width, it is good practice to inspect more than one bin choice before making a strong claim about shape.

A histogram may be unimodal, bimodal, or multimodal. Multiple peaks often suggest subgroups, different data sources, or different operating conditions. The next EDA step is usually to color, facet, or group the data by a relevant categorical variable.

1.6.2 Bivariate Visual EDA

Bivariate EDA examines the relationship between two variables. The plot depends on the type of each variable.

Variable 1	Variable 2	Recommended plot
Categorical / ordinal	Categorical / ordinal	Grouped bar plot
Categorical / ordinal	Numerical	Grouped box plot
Numerical	Categorical / ordinal	Grouped box plot
Numerical	Numerical	Scatterplot

Table 1.2: Recommended bivariate EDA plots by variable-type pairing.

Grouped bar plots compare category combinations, especially counts or proportions. Grouped box plots compare numerical distributions across categories, including medians, IQRs, skewness, and outlier flags. Scatterplots compare two numerical variables and are often the first place we notice whether a linear model is a reasonable first approximation. A visible curve, fan shape, cluster pattern, or outlier should be noted before modeling.

1.6.3 Multivariate Visual EDA

Multivariate EDA examines more than two variables. True high-dimensional visualization is difficult, so practical multivariate EDA often uses combinations of pairwise views.

A **scatterplot matrix** displays many pairwise relationships at once. If there are p numerical variables, there are

$$\frac{p(p-1)}{2} \tag{1.14}$$

unique pairwise scatterplots. The diagonal entries are often histograms or density plots for the individual variables.

A **correlation heatmap** displays many pairwise correlations compactly. It can reveal blocks of related variables, strongly positive or negative relationships, possible multicollinearity before regression, and variables that appear weakly related to all others. A heatmap is not truly multivariate in the sense of showing all variables simultaneously; it summarizes many pairwise linear relationships.

1.7 Applications of Standard Deviation

Standard deviation becomes especially useful when the distribution is approximately bell-shaped. If X is approximately normal with mean μ and standard deviation σ , then approximately

$$\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \quad (1.15)$$

$$\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95, \quad (1.16)$$

$$\mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997. \quad (1.17)$$

This is the **empirical rule**. It is an approximation, not a universal law. It should be used only when the histogram is reasonably bell-shaped or when a normal model is otherwise justified.

When the data are not approximately normal, Chebyshev's inequality provides a more general lower bound. For a random variable X with finite mean μ and finite nonzero variance σ^2 , for any $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}. \quad (1.18)$$

Equivalently, for $k > 1$,

$$\mathbb{P}(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}. \quad (1.19)$$

Window	Empirical rule, if bell-shaped	Chebyshev lower bound
Within 1σ	about 68%	no useful positive bound
Within 2σ	about 95%	at least 75%
Within 3σ	about 99.7%	at least 88.9%

The practical takeaway is simple: if the histogram looks bell-shaped, the empirical rule is more informative; if the shape is unknown or non-normal, Chebyshev is safer but more conservative.

1.8 Covariance, Correlation, and Heatmaps

For two numeric variables X and Y , covariance measures whether they tend to move together. The population covariance is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (1.20)$$

The sample covariance is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}). \quad (1.21)$$

Covariance depends on units, which makes it hard to compare across variable pairs. Correlation rescales covariance into the interval $[-1, 1]$:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \quad (1.22)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}. \quad (1.23)$$

Let

$$\begin{aligned}\tilde{\mathbf{x}} &= \mathbf{x} - \bar{x}\mathbf{1}, \\ \tilde{\mathbf{y}} &= \mathbf{y} - \bar{y}\mathbf{1}.\end{aligned}$$

Then the sample correlation can be written as

$$r_{xy} = \frac{\tilde{\mathbf{x}}^T \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|}. \quad (1.24)$$

This is the cosine of the angle between the centered data vectors. If they point in nearly the same direction, correlation is positive and close to 1. If they point in nearly opposite directions, correlation is negative and close to -1. If they are nearly orthogonal, correlation is near 0. See the Math Foundations textbook, Ch05 Sec. 5.8.

Important caution. Correlation measures linear association. A correlation near 0 does not prove that two variables are unrelated. The relationship may be curved, segmented, or driven by subgroups. Always pair correlation with a scatterplot when possible.

If $\mathbf{X} \in \mathbb{R}^{n \times p}$ contains p numerical variables, then the sample correlation matrix is

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix}, \quad (1.25)$$

where r_{jk} is the sample correlation between variables j and k . For nonconstant variables, the diagonal entries satisfy $r_{jj} = 1$; constant-variable correlations are undefined. The matrix is symmetric because $r_{jk} = r_{kj}$.

1.9 A Practical EDA Workflow

A reusable EDA workflow is:

1. Confirm the variable type and check missingness.
2. Compute center, spread, shape, and extreme summaries for key numerical variables.
3. Make a univariate plot and compare the plot to the summaries.
4. For pairs of variables, choose the plot based on the two variable types.
5. For many numerical variables, inspect a scatterplot matrix or correlation heatmap.
6. Record unusual patterns, possible explanations, and follow-up questions.

The goal is not to produce every possible plot. The goal is to learn enough about the data to make the next modeling or analytic step more defensible.

Coding companion reference. Package-specific Python/R commands for EDA summaries, grouped plots, and heatmaps have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual EDA workflow and interpretation guidance.

1.10 Compact EDA Example

Suppose `pft_score` is a numerical variable. A first EDA pass might compute the mean, median, standard deviation, IQR, minimum, and maximum, then create a histogram. If the mean and median are close and the histogram is roughly symmetric, the empirical rule may be a reasonable approximation. If the histogram is skewed or multimodal, the median and IQR may better represent the typical value and spread.

If `score` is numerical and `gender` is categorical, a grouped box plot compares distributions across groups. Do not stop at comparing medians. Inspect IQRs, outlier flags, skewness within each group, group sample sizes, and overlap between groups. A grouped box plot can reveal a pattern, but it does not explain why the pattern exists.

Finally, suppose a medical data set contains variables such as `triceps`, `insulin`, and `bmi`. If zeros are impossible or implausible for those measurements, then a correlation heatmap before cleaning may be misleading because placeholder zeros create artificial relationships. Recode impossible zeros as missing, document the decision, and create summaries and plots after cleaning.

1.11 Reading EDA Output Defensibly

EDA should produce observations that are specific, cautious, and tied to evidence.

Weak statement	Better EDA statement
The data are normal.	The histogram is roughly bell-shaped and symmetric, so the empirical rule may be reasonable.
There are outliers.	Tukey fences flag three high values; check units and entry errors before removal.
These variables are related.	The scatterplot shows a positive, roughly linear association; the sample correlation is positive.
Group A is better than Group B.	Group A has a higher median and smaller IQR in this sample; this plot alone does not establish causation.
The heatmap proves multicollinearity.	The heatmap suggests strong pairwise correlations; formal modeling diagnostics can check multicollinearity later.

A defensible EDA statement distinguishes what was observed from what was inferred.

1.12 Reusable EDA Checklist

Use this compact checklist when exploring a cleaned data set:

- Identify each variable type: continuous, discrete, ordinal, categorical, date, text, or identifier.
- Check missingness, minimum, maximum, impossible values, and suspicious zeros.
- Choose center and spread summaries that match the variable and distribution.

- Inspect shape: symmetry, skewness, modality, tails, gaps, clusters, and outlier flags.
- For pairs of variables, choose the plot by variable types and look for linear, curved, grouped, or outlier-driven patterns.
- For many numerical variables, use scatterplot matrices or heatmaps to decide which relationships deserve closer inspection.
- Document the evidence, the follow-up question, and the limitation of each claim.

1.13 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 4 use	Math Foundations connection	MF textbook location
A numeric variable as a data vector	$\mathbb{R}^n$ vector notation, entries, and vectors as collections of quantities	Ch02 Secs. 2.1, 2.2, 2.6
Mean as a vector operation	Dot product examples, including $\mathbf{1}^T \mathbf{a} = \sum_i a_i$ and $(1/n)\mathbf{1}^T \mathbf{a}$ as a sample mean	Ch02 Sec. 2.4; Ch05 Secs. 5.1, 5.3
Variance and standard deviation as squared length	Dot products, $\mathbf{a}^T \mathbf{a} = \sum_i a_i^2$, and Euclidean norm $\ \mathbf{x}\ = \sqrt{\mathbf{x}^T \mathbf{x}}$	Ch05 Secs. 5.1, 5.3, 5.5
Centered-vector view of spread	Vector subtraction, scalar multiplication, norm properties, and distance	Ch03 Secs. 3.1, 3.2, 3.4; Ch05 Secs. 5.5, 5.7
Correlation as geometric alignment	Angle formula, dot products, norms, and the orthogonality case $\mathbf{a}^T \mathbf{b} = 0$	Ch05 Secs. 5.1, 5.3, 5.5, 5.8
Scatterplot matrices and heatmaps	Matrix dimensions, submatrix notation, transpose, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.2, 7.4, 7.6.1, 7.6.2

1.14 Chapter Summary

The main ideas from this chapter are:

- Single-number summaries describe center, spread, shape, and extremes.
- The mean is a balance point but is sensitive to outliers; the median and IQR are more robust.
- Skewness measures asymmetry; kurtosis emphasizes tail weight.
- Tukey fences flag possible outliers but do not justify automatic deletion.
- Histograms show univariate numerical structure, including skewness, modality, and approximate bell shape.
- The empirical rule is useful for approximately Gaussian distributions; Chebyshev's inequality gives a conservative bound when normality is not justified.

- Bivariate plots should be selected based on the two variable types.
- Scatterplot matrices and correlation heatmaps summarize many pairwise relationships.
- Correlation is a scaled covariance and can be interpreted geometrically as the cosine between centered data vectors.
- EDA claims should be specific, cautious, and tied to the actual summary or plot that supports them.

The next chapter begins simple linear regression. The EDA habits from this chapter will matter immediately: before fitting a line, we should understand the response distribution, inspect the predictor, check the scatterplot, and decide whether a linear relationship is a reasonable first approximation.